An Agentic Framework Using Slime Mold Intelligence for Transportation Network Optimization

JITENDRA SUNDESHA
jitendrasundesha2006@gmail.com

NIMAL S
24623114@gmail.com

ROSHAN PAUL PATRICK S
24623114@gmail.com

DR. R. ANANDAN
anandan.se@vistas.ac.in

Department of Computer Science and Engineering (AI), Vels Institute of Science, Technology and Advanced Studies (VISTAS), Chennai, India

ABSTRACT Planning a transport network means trading construction cost against travel efficiency and tolerance to link failure. The plasmodium of the slime mold *Physarum polycephalum* has been shown to build networks whose cost, efficiency and fault tolerance are comparable to those of the Tokyo rail system when food sources are placed at city positions. This paper applies a Jones-style multi-agent slime mold simulation to the Chennai Metro. Station coordinates were collected from OpenStreetMap for the Green Line, the Blue Line (Mannadi to the Airport) and the Orange Line (Lighthouse to Poonamallee), mapped onto a two-dimensional grid, and used as nutrient sources for a population of simple agents that deposit and follow a diffusing chemoattractant. The network that emerged was compared with the existing layout by visual overlay. It almost matched the existing layout except at a few locations. The comparison is qualitative, and we state what it can and cannot show. We also set out an evaluation protocol based on normalised total length, mean shortest-path distance and single-link fault tolerance, to be applied in the quantitative stage of the work.

INDEX TERMS *Physarum polycephalum*, slime mold algorithm, transport network design, multi-agent simulation, bio-inspired computation, network optimization, logistics routing, sustainable design.

I. INTRODUCTION

A transport network has to satisfy three goals that pull against each other: the material needed to build the links (total length), the distance travellers actually cover (efficiency), and the ability to keep working when a link fails (fault tolerance). A minimum spanning tree (MST) is the cheapest way to connect a set of points but breaks apart with any single failure, while a fully connected Delaunay triangulation is far more robust and far longer; in the Tokyo study discussed below it was about 4.6 times the MST length [1]. Real networks lie between these extremes, and added redundancy rarely pays for itself in the short term, which is one reason failures of power, rail and information networks keep exposing weak designs [1].

Biological transport systems face the same trade-off without any central planner. Work on biologically inspired network systems repeatedly points to local interaction, redundancy and feedback control as the properties that give such systems their robustness [2]. The slime mold *Physarum polycephalum* is a particularly convenient example. Placed in a maze with food at two exits, the plasmodium retreats from dead ends and leaves a tube along the shortest route [3]. Placed on a template of 36 food sources arranged like the cities around Tokyo, it grew a tube network whose total length relative to the MST was 1.75 ± 0.30, against roughly 1.8 for the real rail network, with the same mean shortest-path ratio (about 0.85) and somewhat lower fault tolerance [1]. The tube network arises from a positive feedback between flux and tube thickness: tubes that carry more flow widen, and tubes that carry little flow shrink [1], [4].

Several computational abstractions of this behaviour exist. The flux–conductance model of Tero *et al.* [4] evolves conductivities on a fixed graph. Jones [5] instead uses a population of particle-like agents that move over a raster image, deposit a chemoattractant and follow its gradient, so no candidate graph is needed beforehand and obstacles can be encoded in the image. Tsompanas *et al.* [6] recast the process as a cellular automaton (CA) and reproduced motorway networks of five regions in minutes rather than the days a laboratory plasmodium needs. We adopt the model of Jones [5] because the input to our problem, a set of points on a map, is naturally a raster, and because the same representation can later accommodate terrain constraints.

In the studies we reviewed, the networks examined are those of Japan, Europe, North America and Mexico; we found none that addresses an Indian transport network. This paper takes a first step in that direction. We build a pipeline that turns hub coordinates into a grid, runs the slime mold simulation, and extracts a network, and we apply it to the Chennai Metro. Our contributions are modest and specific: (i) a working simulation pipeline for generating a slime mold network from geographic hub coordinates; (ii) a Chennai Metro case study with a hand-compiled station dataset and a qualitative comparison between the generated and existing networks;

and (iii) a defined protocol for the quantitative evaluation that will follow.

II. BACKGROUND AND RELATED WORK

A. FLUX–CONDUCTANCE MODELS AND REAL-NETWORK VALIDATION

Tero *et al.* [4] formalised the tube dynamics as an ODE in which the conductivity of each edge grows with the flux it carries and decays at a fixed rate. Their analysis showed that the feedback exponent decides what survives. When the exponent exceeds 1 the outcome depends on the initial conductivities, when it is below 1 all competing paths survive, and when it equals 1 the shortest path always wins, though more slowly [4]. A companion study demonstrated the same solver on a road map of US interstate highways, and showed that a blocked highway segment can be handled by removing the segment, resetting the source to the current position and solving again [7]. The Tokyo experiment [1] extended the idea from paths to multi-point networks and introduced the three normalised measures we reuse here: total length (TL), mean minimum distance between node pairs (MD) and fault tolerance (FT), each relative to the MST. In the same work a sigmoidal flux–conductance model with random source and sink pairs produced networks whose benefit/cost ratio matched or exceeded that of the rail network [1]. These models operate on a given graph, so they rank or prune candidate connections but do not by themselves propose new junctions in open space.

B. SPATIAL AGENT-BASED AND CELLULAR AUTOMATON MODELS

Jones [5] represents the plasmodium as a population of agents on a grid. Each agent senses the chemoattractant trail with three forward sensors, turns toward the strongest signal, steps forward, and deposits chemoattractant if the step succeeds; a blocked agent deposits nothing and picks a random new heading. Because agents both deposit and follow the same substance, nearby agents reinforce one another's trails. The sensor angle (SA), rotation angle (RA) and sensor offset (SO) largely determine whether the population forms reticular, labyrinthine or disconnected patterns, and small offsets of roughly 3 to 7 pixels give fine-grained networks [5]. When fixed attractant sources are projected onto the trail map, the population condenses around them and the network shortens toward a Steiner tree; raising the strength of the projected stimulus removes the Steiner points and leaves an MST-like structure [5]. The same work shows that damaged networks can repair themselves [5].

Tsompanas *et al.* [6] describe a CA in which each cell holds a percentage of Physarum mass and of chemoattractant, both diffusing over a Moore neighbourhood, with mass drawn toward higher attractant. Nutrient sources captured by the mass become new starting points, and tubes are traced along the steepest ascent of the mass field. Compared across five national motorway networks, their model matched 91.3% of the graphs grown by laboratory plasmodia, reproduced the relative neighbourhood graph almost entirely, and shared more than half the edges of the real motorway networks while adding no more than 24% redundant edges [6]. The authors attribute the remaining mismatch with real motorways to terrain, population and political factors that the model does not see [6]. CA models of this family have also been used for crowd evacuation, where building walls are unavailable cells and exits act as food sources [11].

C. COMPUTATIONAL COST

A recurring limitation is run time. Graph-based Physarum solvers must solve a linear system for node pressures at every iteration and may need hundreds of iterations. Zhang *et al.* [8] added an energy term to the basic model and reported fewer iterations and shorter running time than the basic model on random networks of 15 to 2000 nodes. Gao *et al.* [9] pruned inactive nodes and edges and stopped the solver early once near-optimal paths carried most of the flow, testing on random graphs and two real traffic networks. Liu *et al.* [10] applied a Physarum-based algorithm to the Steiner tree problem in networks. These accelerations target graph solvers for shortest-path and Steiner problems; carrying them over to a grid-based agent model would require separate work.

D. POSITIONING OF THIS WORK

Prior validations of Physarum-inspired design rely on laboratory growth or national-scale datasets from outside India. We use the grid-and-agent representation of [5] with a distance-only objective, apply it to the Chennai Metro, and compare it with the existing network. Acceleration, obstacle handling and emergency routing, which the related work shows are feasible in principle [6], [7], [9], [11], are outside the scope of the present results.

III. PROBLEM STATEMENT AND OBJECTIVES

Given the geographic positions of a set of transport hubs, the problem is to generate, by a decentralised bio-inspired process, a network connecting them, and to judge how closely that network resembles an existing one. The objectives are to build a pipeline from hub coordinates to an extracted network, to run it on the Chennai Metro station set, to compare the generated and existing networks, and to fix a quantitative evaluation protocol for the next stage.

IV. METHODOLOGY

A. SYSTEM OVERVIEW

The system has six stages, shown in Fig. 1: geospatial data collection, grid mapping, simulation initialisation, slime mold simulation, network extraction, and evaluation against the as-built network.

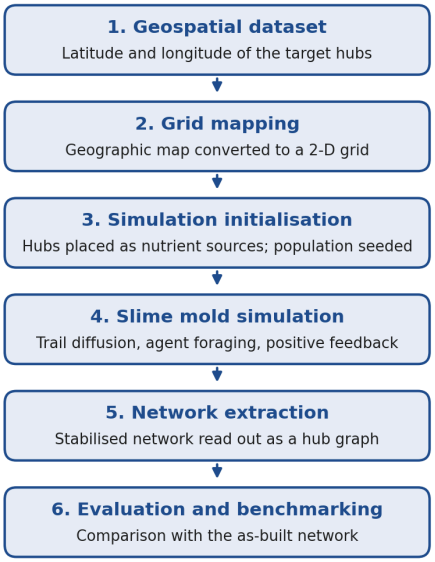


FIGURE 1. Pipeline of the proposed system. Hub coordinates are mapped onto a grid, hubs are placed as nutrient sources, a slime mold simulation grows the network, and the extracted network is compared with the existing infrastructure.

B. STATION DATASET

Station coordinates were collected from OpenStreetMap as latitude–longitude pairs. The dataset covers the full Green Line, the Blue Line from Mannadi to the Airport, and the Orange Line from Lighthouse to Poonamallee. The dataset comprises 56 unique stations across the three lines: 17 on the Green Line, 14 on the Blue Line (Mannadi–Airport), and 28 on the Orange Line (Lighthouse–Poonamallee). The line-wise counts sum to 59 because interchange stations are included in the counts of each line they serve but are counted only once in the total number of unique stations. At the time of data collection, the Green and Blue Lines were fully operational, while the Orange Line remained under construction, with certain sections expected to become operational in the near future. Accordingly, the dataset includes stations from both currently operational and forthcoming sections of the network.

C. GRID MAPPING AND INITIALISATION

Following [5], the environment is represented as a two-dimensional raster in which cells correspond to habitable regions, obstacles, and nutrient sources. The geographic coordinates of the stations were mapped to grid coordinates using linear min–max normalization of their latitude and longitude values, preserving their relative spatial positions within the simulation domain. Each station was then assigned to the corresponding grid cell and treated as a nutrient source, whose stimulus was projected onto the chemoattractant trail map following the pre-patterning approach described in [5]. The simulation was initialised with the agent population around the centre of the mapped region. In the present experiments, the grid was treated as a uniform plane without obstacles.

D. SLIME MOLD MODEL

The model follows Jones [5]. Each agent occupies one grid cell and keeps a floating-point position and heading. At each scheduler step an agent samples the trail map at three forward sensors, placed at the sensor offset SO and at angles of plus and minus SA from its heading, rotates by RA toward the strongest reading, and attempts to move one cell. A successful move deposits a fixed amount of chemoattractant; a blocked agent deposits nothing and takes a random heading. After every step the trail map is smoothed with a 3×3 mean filter and decayed. Agents are processed in random order to avoid ordering bias. Because deposit and sensing use the same substance, trails attract further agents, and the positive feedback that thickens busy paths and starves unused ones emerges from the population rather than from an explicit equation. Table 2 lists the parameters; the typical values from [5] are given for reference, and the settings used here must be filled in.

TABLE 2. Model parameters.

Parameter	Meaning	Typical in [5]	Used here
SA	Sensor angle	22.5° or 45°	45°
RA	Rotation angle	45°	45°
SO	Sensor offset	9 px	8 px
depT	Deposit per step	5	10
decayT	Trail decay	0.1	0.1
diffK	Kernel size	3	3
%p	Population (% of grid)	3–15	15
wProj	Station stimulus weight	0.01–0.1	0.05

E. NETWORK EXTRACTION

The simulation produces a spatial field representing the distribution of agents and chemoattractant concentration rather than a graph directly. After the simulation reaches a stable state, the chemoattractant field is thresholded to identify regions corresponding to persistent transport paths. These regions are then converted into a network representation by extracting the connected paths between nutrient-source stations. The resulting network is represented as a set of links connecting the corresponding stations and is used for comparison with the existing Chennai Metro network, as shown in Fig. 3. The simulation was run for a fixed number of iterations, after which the resulting chemoattractant field was used for network extraction. The exact threshold and stopping criterion were kept consistent across experiments.

F. EVALUATION PROTOCOL

The comparison reported in this paper is a visual overlay of the generated and existing networks. For the quantitative stage we adopt the three measures of [1], each normalised by the MST over the same stations. TL is the total link length. MD is the mean shortest-path distance over all pairs of stations. FT is the fraction of single-link removals that disconnect part of the network, which equals 1 for the MST because every link of a tree is a bridge. TL/TL_{MST} then measures cost, MD/MD_{MST} measures efficiency (values below 1 indicate shorter journeys than on the MST), and FT/FT_{MST} measures resilience. We also plan to compute edge overlap between

the generated and real networks, in the manner of [6], and to include the MST, the relative neighbourhood graph and the Delaunay triangulation of the same stations as reference graphs.

V. EXPERIMENTAL SETUP

The simulation was implemented in Python 3.14.7 and executed on a GNU/Linux system equipped with a 12th-generation Intel Core i5 processor and 16 GB of RAM. The experiment was performed over 10 independent runs, with each run consisting of 1,000 scheduler steps and using a different random initialisation of agent positions and headings. The model is stochastic because the initial agent positions and headings are randomly assigned [5], and consequently, individual runs may produce slightly different network structures. The network shown in Fig. 3 represents a representative run selected from the 10 simulations.

VI. RESULTS

A. NETWORK GROWTH

Fig. 2 is a frame from a run in progress. The population has spread from the central region of the grid and covers the stations in the middle of the map, while stations along the western and eastern arms and at the southern end are still isolated points. Growth from the centre outward toward the peripheral stations is consistent with the foraging stage described for the real organism [1].

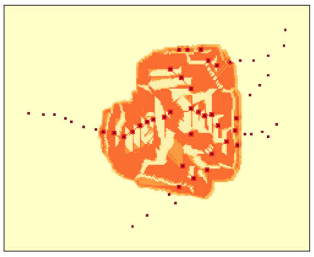
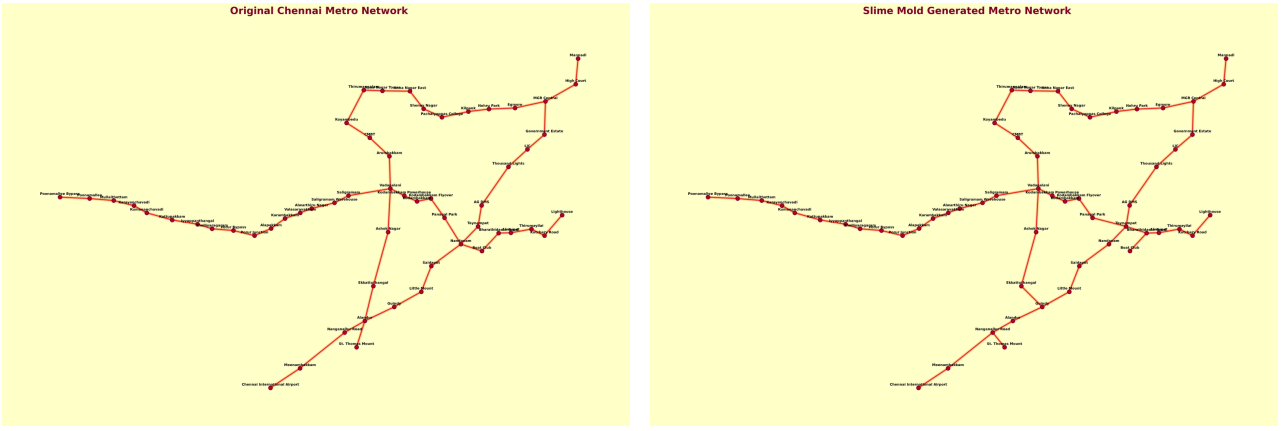


FIGURE 2. Intermediate stage of a simulation run. Dark red points are stations acting as nutrient sources; orange regions are occupied by the agent population.



(a) (b)

FIGURE 3. Chennai Metro network for the same set of stations: (a) existing network, (b) network generated by the slime mold simulation. The two differ at three locations.

B. COMPARISON WITH THE EXISTING NETWORK

Fig. 3 places the existing Chennai Metro network alongside the network generated by the simulation. The two layouts show a high degree of visual similarity, particularly in the overall west–east corridor and the major branches toward the northern and southern sections of the network. However, the generated network differs from the existing network at three locations. First, the model connects St. Thomas Mount to Nanganallur Road instead of forming the connection through Alandur. Second, it connects Ekkattuthangal to Guindy rather than routing the connection through Alandur. Third, the model places the interchange at Teynampet instead of Nandanam, resulting in a different interchange configuration in this section of the network.

C. QUANTITATIVE MEASURES

A quantitative comparison of the existing and generated networks is planned using total link length (TL), mean shortest-path distance (MD), and single-link fault tolerance (FT), each normalised with respect to the corresponding minimum spanning tree (MST). These measures will provide quantitative estimates of network cost, travel efficiency, and resilience. The present study focuses on the qualitative comparison of the generated and existing networks; therefore, quantitative evaluation is left for future work.

VII. DISCUSSION

The simulation, using only the positions of stations, produced a network close to the existing Chennai Metro. This is in line with earlier findings that slime mold networks approximate real infrastructure in Tokyo [1] and in several motorway systems [6]. It does not show that the Chennai alignment is optimal, or that the simulation

would predict a new alignment; it shows that the layout is close to what a purely distance-driven decentralised process proposes.

Two points need care. First, metro lines are close to chains of stations, and many reasonable distance-based procedures would connect neighbouring stations of a chain in the same way. A visual match on such a dataset is therefore a weaker test than a match on the scattered cities of the Tokyo or motorway studies. The reference graphs listed in Section IV-F, and the three normalised measures, are needed to show how much the slime mold result adds over simpler baselines. Second, we think the three differences may come from the fact that the current implementation optimises distance only. A real alignment is shaped by demand, land use, existing roads and political decisions, which the model does not see; Tsompanas *et al.* make a similar argument about motorways [6]. We have not tested this explanation.

The extra cross-links in the generated network resemble the redundant links that Physarum networks add to improve efficiency and resilience [1]. In the model of Jones [5], the strength of the projected station stimulus controls how tight the network is: a stronger stimulus removes redundant links, a weaker one keeps them. If the cross-links in our output come from this mechanism, the parameter wProj offers a way to tune the redundancy of the generated network, but this requires a parameter study that has not yet been done.

The approach has practical appeal as a planning aid. It needs only coordinates, runs without a pre-specified graph, and its representation, a raster image, can encode terrain and barriers in later versions [5], [6]. Whether it adds value for a planner depends on the quantitative comparison to come.

VIII. LIMITATIONS

The evidence is qualitative and rests on one case, one parameter setting and 10 independant runs. No metrics were computed, no baselines were compared, and the sensitivity of the output to SA, RA, SO, population density or projection weight was not studied, although [5] shows these parameters strongly affect pattern formation. Station coordinates were entered by hand and have not been cross-checked against an official dataset. The model optimises distance only and treats the map as a uniform plane. The extraction step that turns the simulated field into a graph is a design choice that can affect which links appear. The work reported here does not implement the humanitarian-logistics and emergency-response features mentioned in the project proposal.

IX. CONCLUSION AND FUTURE WORK

We built a pipeline that turns hub coordinates into a grid, runs a Jones-style multi-agent slime mold simulation, and extracts a network, and we applied it to a hand-compiled Chennai Metro station set. The generated network was visually close to the existing one, differing in three places. Because the comparison is visual and the dataset is chain-like, the result is best read as an encouraging first check rather than as a validation.

The next steps follow from the limitations. First, compute the normalised TL, MD and FT measures and edge overlap for the generated, existing and baseline networks, over repeated runs. Second, carry out a parameter study of SA, RA, SO, population and projection weight. Third, add terrain and barriers through the image environment, as supported in [5] and anticipated in [6]. Fourth, apply the pipeline to other Indian networks such as freight corridors and regional rapid transit. Fifth, improve the speed of the simulation, drawing on the pruning and early-termination ideas of [9] for graph solvers and on the CA formulation of [6]; the planned migration of the engine to Rust belongs here. Last, the emergency-response direction suggested by [7] and [11], in which blocked links or hazards force the network to re-route, is a natural application once obstacles and quantitative evaluation are in place.

REFERENCES

[1]A. Tero, S. Takagi, T. Saigusa, K. Ito, D. P. Bebber, M. D. Fricker, K. Yumiki, R. Kobayashi, and T. Nakagaki, “Rules for biologically inspired adaptive network design,” *Science*, vol. 327, no. 5964, pp. 439–442, Jan. 2010.

[2]T. Nakano, “Biologically inspired network systems: A review and future prospects,” *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.*, vol. 41, no. 5, pp. 630–643, Sep. 2011.

[3]T. Nakagaki, H. Yamada, and Á. Tóth, “Intelligence: Maze-solving by an amoeboid organism,” *Nature*, vol. 407, no. 6803, p. 470, 2000.

[4]A. Tero, R. Kobayashi, and T. Nakagaki, “A mathematical model for adaptive transport network in path finding by true slime mold,” *J. Theor. Biol.*, vol. 244, no. 4, pp. 553–564, 2007.

[5]J. Jones, “Characteristics of pattern formation and evolution in approximations of Physarum transport networks,” *Artif. Life*, vol. 16, no. 2, pp. 127–153, 2010.

[6]M.-A. I. Tsompanas, G. Ch. Sirakoulis, and A. I. Adamatzky, “Evolving transport networks with cellular automata models inspired by slime mould,” *IEEE Trans. Cybern.*, vol. 45, no. 9, pp. 1887–1899, Sep. 2015.

[7]A. Tero, R. Kobayashi, and T. Nakagaki, “Physarum solver: A biologically inspired method of road-network navigation,” *Physica A*, vol. 363, pp. 115–119, 2006.

[8]X. Zhang, Q. Wang, A. Adamatzky, F. T. S. Chan, S. Mahadevan, and Y. Deng, “An improved *Physarum polycephalum* algorithm for the shortest path problem,” *Sci. World J.*, vol. 2014, Art. no. 487069, 2014.

[9]C. Gao, X. Zhang, Z. Yue, and D. Wei, “An accelerated Physarum solver for network optimization,” *IEEE Trans. Cybern.*, vol. 50, no. 2, pp. 765–776, 2020, doi: 10.1109/TCYB.2018.2872808.

[10]L. Liu, Y. Song, H. Zhang, H. Ma, and A. V. Vasilakos, “Physarum optimization: A biology-inspired algorithm for the Steiner tree problem in networks,” *IEEE Trans. Comput., vol. 64, no. 3, pp. 819-832,2015, doi: 10.1109/TC.2013.229.*

[11]V. S. Kalogeiton, D. P. Papadopoulos, I. P. Georgilas, G. Ch. Sirakoulis, and A. I. Adamatzky, “Cellular automaton model of crowd evacuation inspired by slime mould,” *Int. J. Gen. Syst.*, vol. 44, no. 3, pp. 354–391, 2015.